6.3.3 Models for Heat Baths and Langevin's Model for Brownian Motion

If the Ornstein-Uhlenbeck SDE models the momentum of particles undergoing collisions, it can be complemented by a corresponding differential equation for the position of the form

$$\mathrm{d}q = m^{-1} p \mathrm{d}t,$$

to provide a full description of the motion. This is the key idea in Langevin's model for Brownian motion [213].

Let us derive the Langevin dynamics model for the special case of a heavy particle (mass one) interacting with a large collection of light particles. We assume as in [138] that we are given a system of k coupled oscillators (a "heat bath") in contact with a distinguished particle. The Hamiltonian is

$$H = \frac{P^2}{2} + \sum_{i=1}^{k} \frac{p_i^2}{2\mu_i} + U(Q) + \frac{1}{2k} \sum_{i=1}^{k} (q_i - Q)^2$$

where Q and P represent the position and momentum of the distinguished particle and $\{q_i, p_i\}$ are the positions and momenta of the particles of the bath, μ_i giving the mass of the ith light particle.

The equations of motion for the combined system are

$$\dot{Q} = P, \tag{6.22}$$

$$\dot{P} = -U'(Q) - k^{-1} \sum_{i=1}^{k} (Q - q_i), \tag{6.23}$$

where, for $i = 1, 2, \ldots, k$,

$$\dot{q}_i = p_i/\mu_i, \tag{6.24}$$

$$\dot{p}_i = (Q - q_i)/k. \tag{6.25}$$

Because the system (6.24)–(6.25) is linear, we may solve it analytically for $\{q_i, p_i\}$; this yields:

$$q_i(t) = \cos(\Omega_i t) q_i(0) + \Omega_i^{-1} \sin(\Omega_i t) p_i(0)/\mu_i + k^{-1} \mu_i^{-1} \int_0^t \Omega_i^{-1} \sin(\Omega_i (t-s)) Q(s) \mathrm{d}s, \tag{6.26}$$